FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE (Advanced), 2015

PART TEST – III

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 08)** contains 8 multiple choice questions which have one or more correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (09 to 16) contains 4 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (17 – 20) contains 4 Matching Lists Type questions: Each question has four statements in LIST I & 4 statements in LIST II. The codes for lists have choices (A), (B), (C), (D) out of which only one is correct. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

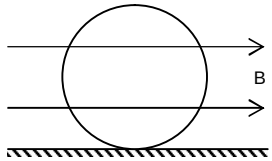
PART – I

SECTION – A

Single Correct Choice Type

*This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

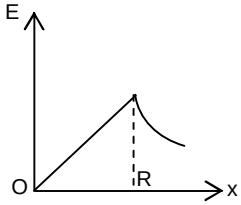
1. A conducting ring of mass 2 kg and radius 0.5 m is placed on a smooth horizontal plane. The ring carries a current of I = 4A. A horizontal magnetic field B = 10T, coplanar with ring exist in the region. if the ring is released from rest then



(A) angular acceleration of the ring is 40π rad s^{-2} (B) torque on the ring is $20\ \pi$Nm

(C) angular acceleration of the ring is 20π rad s^{-2} (D) torque on the ring is $10\ \pi$Nm

2. A sphere has a positive charge. Figure shows variation of electric field (E) with distance x from its centre. From this figure, we conclude that



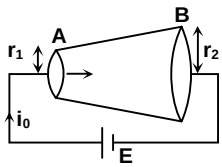
(A) Sphere is made of a non conducting materials
(B) Diameter of sphere is equal to R/2
(C) Electric potential, due to sphere, is maximum at its centre
(D) Density of charge is uniform throughout the volume of sphere

3. Electric charge +q is uniformly distributed over the entire length of a ring of radius r rotating with constant angular velocity ω about its own axis. Assuming mass of ring to be m, its magnetic moment to be M and angular momentum to be L.

(A) $M = \frac{q\omega r^2}{2}$ (B) $M = \frac{2q\omega r^2}{5}$

(C) $\frac{M}{L} = \frac{q}{2m}$ (D) $\frac{M}{L} = \frac{2q}{m}$

4. A battery of emf E is connected across a conductor as shown as one observe from A to B. then

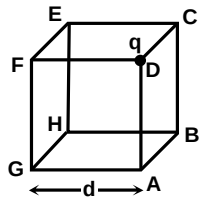


(A) Current remains same
(B) Drift velocity of electron decreases
(C) Electric field decreases
(D) Potential drop across the length decreases

Space for rough work

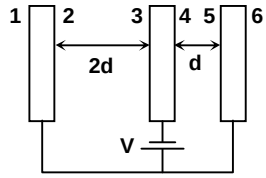
5. To convert galvanometer into a voltmeter, a large resistance is connected in series with the galvanometer. Due to the use of this large resistance is
(A) Accuracy increases (B) Least count increases
(C) Least count decreases (D) Range increase

6. A point charge q is placed at corner D of a cube of side d. Flux of electric field passing through face
(A) ABCD is zero (B) CDFE is zero
(C) DFGA is zero (D) ABHG is $\frac{q}{24\varepsilon_0}$



7. Select the correct statement(s) from the following.
The path of a charged particle moving with an initial velocity u
(A) parallel to the uniform electrostatic field is a straight line.
(B) perpendicular to the uniform electrostatic field is a parabola.
(C) parallel to the uniform magnetic field is a helix with its axis parallel to the direction of B.
(D) perpendicular to the uniform magnetic field is a circle in a plane normal to the field.

8. In the figure shown, area of each plate is A. Match the following
(A) Charge on plate 5 is $\frac{\varepsilon_0 A}{d}V$
(B) Potential difference between plates 2 and 3 is V
(C) Potential difference between plates 2 and 5 is zero
(D) Charge on the plate 3 is zero.

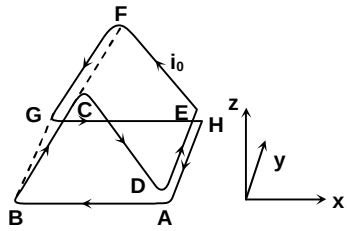


Linked Comprehension Type

*This section contains 4 paragraphs. Based upon each paragraph, 2 multiple choice questions have to be answered. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

Paragraph for Question Nos. 9 to 10

A coil (ABCDEFGH) is in the form of prism having each side equal to length ℓ carries a current i_0 (Assume gap between parallel wires AH and DE to be negligibly small)



9. The direction of magnetic field at the geometric centre of three dimensional prism shaped coil will be
(A) along y-axis (B) along x-axis
(C) along z-axis (D) none of these

Space for rough work

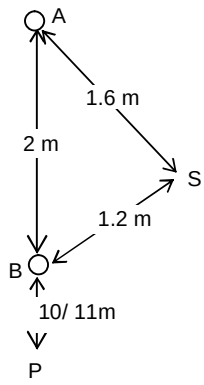
10. If an external magnetic field B_0 is applied along +ve z-axis then the force on the coil ABCDEFGH due to B_0 only will be equal to

(A) zero (B) $L_0B\ell$

(C) $6L_0B\ell$ (D) $2L_0B\ell$

Paragraph for Question Nos. 11 to 12

Two long straight parallel wires are 2 metres part, perpendicular to the plane of the paper. The wire A carries a current of 9.6 amps, directed into the plane of the paper. The wire B carries a current such that the magnetic field of induction at the point P, at a distance of $\frac{10}{11}$ metre from the wire B, is zero. Answer the following questions.



11. The magnitude of direction of the current in B.

(A) 4.0 A (B) 3.0 A

(C) 2.0 A (D) 1.0 A

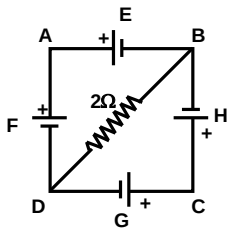
12. The magnitude of the magnetic field of induction at the point S.

(A) 1.1 μT (B) 1.3 μT

(C) 1.4 μT (D) 2.8 μT

Paragraph for Question Nos. 13 to 14

In the circuit shown in fig. E, F, G and H are cells of emf 2, 1, 3 and 1 Volts respectively and their internal resistance are 2, 1, 3 and 1Ω respectively. Calculate



13. The potential difference between B and D and

(A) $\frac{2}{13}$V (B) $\frac{4}{13}$V

(C) $\frac{6}{13}$V (D) $\frac{8}{13}$V

Space for rough work

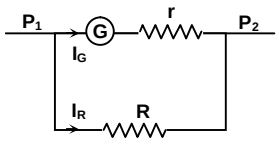
14. The potential difference across the terminals of the cell G.

(A) $\frac{19}{13}$ V (B) $\frac{17}{13}$ V

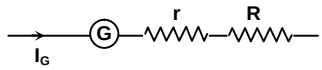
(C) $\frac{21}{13}$ V (D) $\frac{23}{13}$ V

Paragraph for Question Nos. 15 to 16

Measurements and interpretations of voltage and electric current signals are common in modern medicine. Occasionally, a situation arises in which a voltmeter or an ammeter is needed but it is not available. A galvanometer is an instrument that can be used to construct an ammeter. It can also be used to construct a voltmeter. In both cases, a resistor R must be connected to galvanometer to effect the change.



(i)



(ii)

To turn the galvanometer into an ammeter, the resister R is connected in parallel, as shown in the figure (i), The resistor R is connected in series with the galvanometer in order to turn it into a voltmeter, as shown in the figure (ii). The current required to produce a full scale deflection in a galvanometer is 10 mA. The internal resistance of galvanometer is 100 Ω. Let V_r be the voltage across r and V_r the voltage across R.

15. Which of the following formulas correctly applies to the ammeter concept?

(A) $V_r > V_R$ (B) $V_r < V_R$

(C) $V_r = V_R$ (D) more information needed

16. What resistance must be connected in parallel to the galvanometer to turn it into an ammeter capable of reading electric current upto 10.01 A?

(A) 0.1 Ω (B) 1 Ω

(C) 10 Ω (D) none of these

Space for rough work

SECTION – A

Matching List Type

This section contains **4 multiple choice questions**. Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

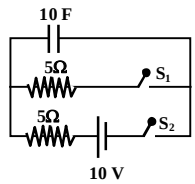
17. List–I specifies a point P at distance r from the center/axis of a symmetrical distribution of charge. List–II gives the variation of electric intensity at P as a function of r(>0). Match the entries of List–I with possible entries of List–II.

List – I		List - II	
(P)	P lies outside a long cylinder having uniform volume charge density	(1)	$E \propto \frac{1}{r^2}$
(Q)	P lies inside a spherical charged conductor	(2)	$E \propto \frac{1}{r}$
(R)	P lies inside a spherical body having uniform volume charge density	(3)	$E \propto r$
(S)	P lies outside of a nonconducting solid sphere of radius R and its volume charge density varies as $\rho = \frac{\rho_0 x^2}{R^2}$, where x is distance from centre of the sphere	(4)	$E \propto r^0$

Codes:

	P	Q	R	S
(A)	2	4	3	1
(B)	4	1	3	2
(C)	2	3	2	4
(D)	4	1	2	3

18. In the circuit shown the capacitor has an initial charge q = 100 C and a capacitor C = 10 F. The emfs of the batteries are shown in the figure and the batteries have negligible internal resistance. The switches are closed at t = 0. Match the following. The quantities on the right columns are given in SI units.



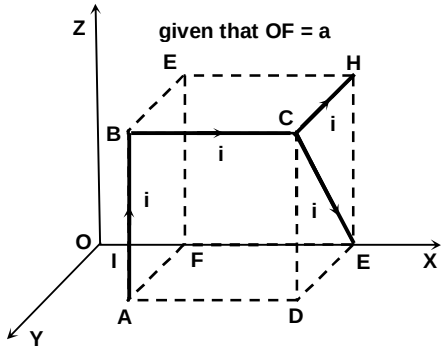
List – I		List - II	
(P)	Time constant of the circuit	(1)	0
(Q)	Current through S_1 at t = 0	(2)	25
(R)	Current through S_2 when $t \to 0$	(3)	2
(S)	Final charge on the capacitor	(4)	50

Codes:

	P	Q	R	S
(A)	3	1	4	2
(B)	2	1	3	4
(C)	4	3	2	1
(D)	2	1	4	3

Space for rough work

19. Side of the cube is 'a' and wire AB, BC, CH and CE are insulated with each other, and each having current I, then match the following.
External magnetic field applied along positive x-axis as $\vec{B} = B_0 x \hat{i}$



	List – I Magnitude of force due to external magnetic field only		List – II	
(P)	On AB	(1)	zero
(Q)	On BC	(2)	$2\sqrt{2}iB_0a^2$
(R)	On CH	(3)	iB_0a^2
(S)	On CE	(4)	$2iB_0a^2$

Codes:

	P	Q	R	S
(A)	3	1	4	2
(B)	2	1	3	4
(C)	4	3	2	1
(D)	1	4	2	3

20. Match the following:

	List – I		List – II
(P)	Self–energy of a uniformly charged spherical shell	(1)	$\frac{1}{8\pi\varepsilon_0}\frac{Q^2}{R}$
(Q)	Energy stored per unit volume of a charged air filled capacitor.	(2)	$\frac{1}{4\pi\varepsilon_0}\frac{2Q^2}{R^2}$
(R)	Force between two circular plates of a charged parallel plate capacitor, radius of the plates being R and separation R/32.	(3)	$\frac{1}{4\pi\varepsilon_0}\frac{Q^2}{R}$
(S)	Work done by electric field in moving a point charge Q from the surface of a uniformly charged sphere of radius R, and moving it to infinity, total charge on the sphere = Q.	(4)	$\frac{1}{2}\varepsilon_0 E^2$

Codes:

	P	Q	R	S
(A)	3	1	4	2
(B)	2	1	3	4
(C)	1	4	2	3
(D)	4	1	2	3

Space for rough work

Chemistry

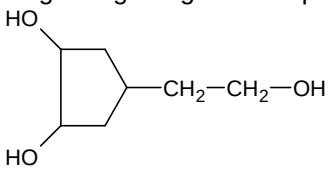
PART - II

SECTION – A

Single Correct Choice Type

*This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

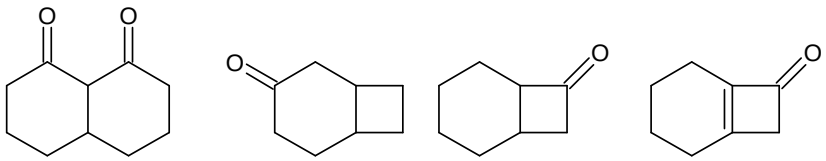
1. Regarding the given compound:



Choose the correct option (s)

(A) The above compound exhibit geometrical as well as optical isomerism
(B) Total number of stereoisomers possible for above compound is 4
(C) One of isomers of the above compounds shows plane of symmetry as well as axis of symmetry
(D) The above compound does not exhibit mesomerism

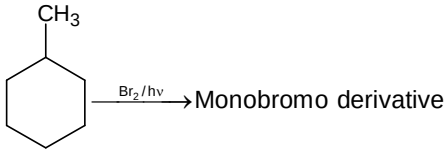
2. In the given compound choose the correct statement (s) regarding percentage enol content in H_2O



(I) (II) (III) (IV)

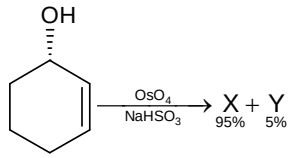
(A) I > III
(B) I > III > IV
(C) II > III
(D) IV > I > II

3. Which of the following option(s) is/are correct regarding given chemical reaction?

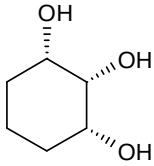
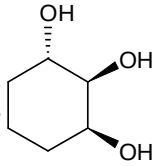


(A) Number of possible mono bromo product (including stereo) is 7
(B) Number of possible mono bromo product (excluding stereo) is 5
(C) Total number of optical active isomers per mono bromo products are 4
(D) Total number of optical inactive isomers for mono product are 2

Space for rough work

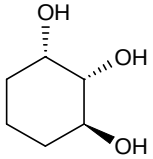
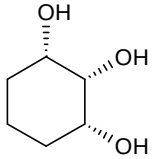
4. 

Which of the following statements are correct regarding product of above reaction?

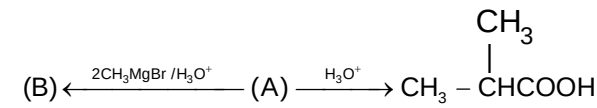
(A) Compound Y is  and compound X is 

(B) Compound X and Y are enantiomers

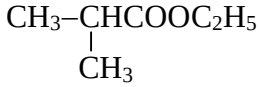
(C) Compound X and Y both are optically active

(D) Compound X is  and compound Y is 

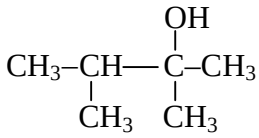
5. Consider the following reaction sequence

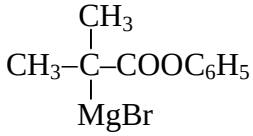


Which of the following statements are true?

(A) Compound (A) can be 

(B) Compound (A) can be 

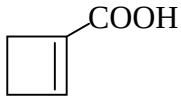
(C) Compound (B) can be 

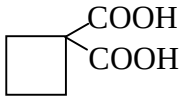
(D) Compound (B) can be 

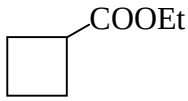
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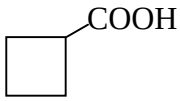
6. In the reaction sequence,

$CH_2(COOEt)_2 + BrCH_2–CH_2CH_2Br \xrightarrow[\text{excess}]{NaOEt} (A) \xrightarrow[\Delta]{H^+/H_2O}$? The possible final product(s) is/are

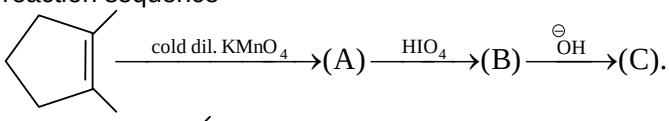
(A) 

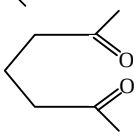
(B) 

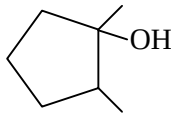
(C) 

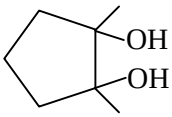
(D) 

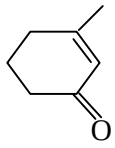
7. In a reaction sequence



(A) (A) is 

(B) (B) is 

(C) (A) is 

(D) (C) is 

8. Which of the following isomer is an example of synthetic rubber?

(A) Buna–N (B) ABS

(C) Neoprene (D) cis–polyisoprene

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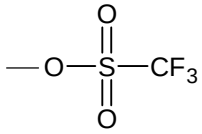
Linked Comprehension Type

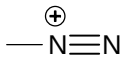
*This section contains 4 paragraphs. Based upon each paragraph, 2 multiple choice questions have to be answered. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

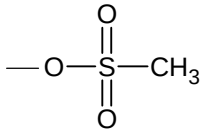
Paragraph for Question Nos. 09 to 10

Nucleophilic aliphatic substitution reaction is given by those compounds which have nucleophilic groups as leaving group. Less is the basicity of the leaving group, more is its leaving power. Leaving group leaves in the form of nucleophile (charged or neutral).

9. Leaving power of which group is maximum?

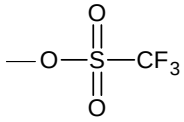
(A) 

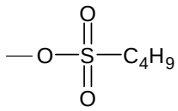
(B) 

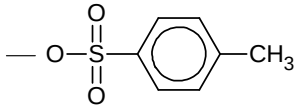
(C) 

(D) 

10. Leaving power of which group is maximum?

(A) 

(B) 

(C) 

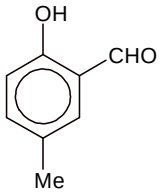
(D) 

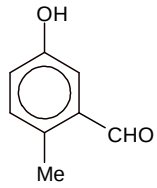
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Paragraph for Question Nos. 11 to 12

Reimer-Tiemann reaction involves an aryl carbanion/enolate anion and also $-CCl_3$ derived from the action of strong bases on $CHCl_3$, though the latter has only a transient existence decomposing to $:CCl_2$, a highly electron deficient electrophile that attacks the aromatic nucleus; the product from phenoxide ion is after acidification, very largely the *o*-aldehyde plus just a small amount of *p*-isomer.

11. If the above reaction is carried out on the anion of *p*-hydroxytoluene, what will be theproduct/products?

(A) 

(B) 

(C) Both (A) and (B)

(D) None of the above

12. Name the electrophile attacking carbanion in the above reaction.
(A) Chloroform
(B) Trichloromethyl anion
(C) dichlorocarbene
(D) hydrogen ion

Paragraph for Question Nos. 13 to 14

Aldehyde having no α-hydrogen when treated with strong base, undergoes redox reaction. This reaction is known as Cannizzaro reaction. Tischenko reaction is also a redox reaction which takes place in the presence of aluminium ethoxide.

13. Which of these compounds will give Cannizzaro reaction?
(A) $C_6H_5CH_2CHO$
(B) CHO–CHO
(C) $C_6H_5CH_2CHO$
(D) All of these

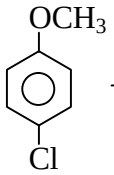
14. Internal crossed Cannizzaro reaction is given by:
(A) C_6H_5COCHO
(B) C_6H_5CHO
(C) CH_3–CHO
(D) C_6H_5–CH_2CO–CHO

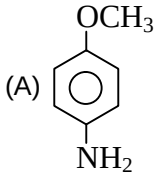
Paragraph for Question Nos. 15 to 16

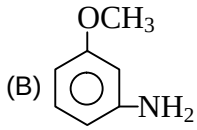
Aryl and vinyl halides are less reactive towards nucleophilic substitution reactions. Chlorobenzene itself and other aryl halides having electron–donating substituents undergo nucleophilic substitution reactions by benzyne mechanism while aryl halides having electron – withdrawing substituents at o- and p-positions undergo bimolecular nucleophilic substitution reactions.
Based on the given passage, reply the following questions:

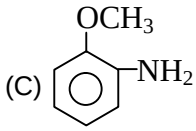
15. In the reaction of p-chlorotoluene with KNH_2 in liq NH_3, the major product is
(A) o–toluidine
(B) m–toluidine
(C) p–toluidine
(D) p–chloroaniline

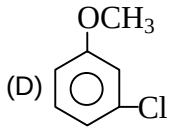
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16. In the reaction  A. The major product is

(A) 

(B) 

(C) 

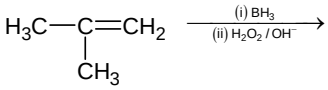
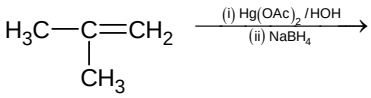
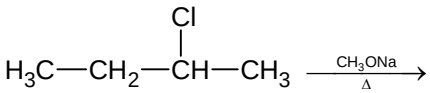
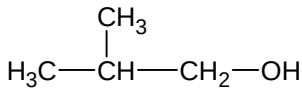
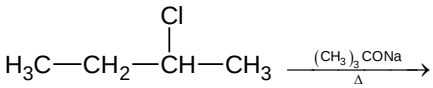
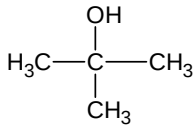
(D) 

SECTION – A

Matching List Type

This section contains **4 multiple choice questions**. Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

17. Match the List – I with List – II:

List – I		List – II	
(P)		(1)	$CH_3 - CH_2 - CH = CH_2$
(Q)		(2)	$CH_3 - CH = CH - CH_3$
(R)		(3)	
(S)		(4)	

	P	Q	R	S
(A)	3	4	2	1
(B)	2	3	4	1
(C)	4	1	2	3
(D)	1	2	3	4

Space for rough work

18. Match the following:

List – I		List – II	
(P)	$RCONH_2 \xrightarrow{Br_2/KOH} RNH_2$	(1)	Curtius rearrangement
(Q)	$RCOOH \xrightarrow[H_2SO_4]{N_3H} R-NH_2$	(2)	Lossen rearrangement
(R)	$RCOCl \xrightarrow[\Delta]{NaN_3} R-NH_2$	(3)	Schmidt rearrangement
(S)	$R-\overset{O}{\overset{\|\|}{C}}-Cl \xrightarrow[OH^-/\Delta]{(i)\ NH_2OH}$	(4)	Hoffmann bromamide degradation

	P	Q	R	S
(A)	3	2	1	4
(B)	2	3	4	1
(C)	4	3	1	2
(D)	1	2	3	4

19. Match the following List – I with List – II:

List – I (Reaction)		List – II (Reagent)	
(P)	$CH_3-CH=CH_2 \longrightarrow CH_3-\overset{Br}{\overset{\|}{C}}H-CH_3$	(1)	HBr
(Q)	$CH_3-CH=CH_2 \longrightarrow CH_3-CH_2-CH_2Br$	(2)	NBS
(R)	$CH_3-CH=CH_2 \longrightarrow BrCH_2-CH=CH_2$	(3)	Br_2/CCl_4
(S)	$CH_3-CH=CH_2 \longrightarrow CH_3-CHBr-CH_2Br$	(4)	HBr (Peroxide)

	P	Q	R	S
(A)	3	2	1	4
(B)	2	3	4	1
(C)	4	1	2	3
(D)	1	4	2	3

Space for rough work

20. Match the following List – I with List – II:

List – I (Substrate + reagent)		List – II (Product)	
(P)	$CH_3–CH_2–CH_2–Cl \xrightarrow[C_2H_5OH]{C_2H_5ONa}$	(1)	$CH_3–CH_2–CH=CH_2$
(Q)	$CH_3–\underset{\displaystyle Cl}{\overset{\displaystyle CH_3}{\overset{\vert}{\underset{\vert}{C}}}}–CH_3 \xrightarrow[C_2H_5OH]{C_2H_5ONa}$	(2)	$CH_3–CH=CH–CH_3$
(R)	$CH_3-CH_2-\underset{\displaystyle Cl}{\overset{\vert}{CH}}-CH_3 \xrightarrow[BuOH]{tBuOK}$	(3)	$CH_3–CH=CH_2$
(S)	$CH_3-CH_2-\underset{\displaystyle Cl}{\overset{\vert}{CH}}-CH_3 \xrightarrow[C_2H_5OH]{C_2H_5ONa}$	(4)	$CH_3–\underset{\displaystyle CH_3}{\overset{\vert}{C}}=CH_2$

	P	Q	R	S
(A)	3	4	1	2
(B)	2	3	4	1
(C)	4	1	2	3
(D)	1	2	3	4

Space for rough work

Mathematics PART – III

SECTION – A

Single Correct Choice Type

*This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

1. The curve $y = ax^2 + bx + c$ passes through the point (1, 2) and its tangent at origin is the line $y = x$. The area bounded by the curve, the ordinate of the curve at minima and the tangent line is

(A) $\frac{1}{24}$ (B) $\frac{1}{12}$

(C) $\frac{1}{8}$ (D) $\frac{1}{6}$

2. Primitive of $I = \int \sqrt{1 + 2\,\text{cosec}\,x \cot x + 2\cot^2 x}\,dx$ w.r.t. x is

(A) $2\ln\cos\frac{x}{2} + c$ (B) $2\ln\sin\frac{x}{2} + c$

(C) $\frac{1}{2}\ln\cos\frac{x}{2} + c$ (D) $\ln \sin x - \ln (\text{cosec } x) + c$

3. A non-zero polynomial with real coefficients has the property that $f(x) = f'(x)\cdot f''(x)$, then the leading coefficient of f(x) is

(A) $\frac{1}{6}$ (B) $\frac{1}{9}$

(C) $\frac{1}{12}$ (D) $\frac{1}{18}$

4. Let $f(x) = \frac{e^{\tan x} - e^x + \ln(\sec x + \tan x) - x}{\tan x - x}$ be a continuous function at x = 0. The value of f(0) equals

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$

(C) $\frac{3}{2}$ (D) 2

Space for rough work

5. For the function $y = f(x) = (x^2 + bx + c)e^x$ which of the following holds?
(A) if $f(x) > 0$ for all real $x \Rightarrow f'(x) > 0$ (B) if $f(x) > 0$ for all real $x \Rightarrow f'(x) > 0$
(C) if $f'(x) > 0$ for all real $x \Rightarrow f(x) > 0$ (D) if $f'(x) > 0$ for all real $x \Rightarrow f(x) > 0$

6. Suppose f is defined from $R \to [-1, 1]$ as $f(x) = \frac{x^2 - 1}{x^2 + 1}$ where R is the set of real number. Then the statement which does not hold is
(A) f is many one onto
(B) f increases for $x > 0$ and decreases for $x < 0$
(C) minimum value is not attained even though f is bounded
(D) the area included by the curve $y = f(x)$ and the line $y = 1$ is π sq. unit

7. People living at mars, instead of the usual definition of derivative D f(x) define a new kind of derivative, D* f(x) by the formula $D^*f(x) = \lim_{h\to 0} \frac{f^2(x+h) - f^2(x)}{h}$ where $f^2(x)$ means $[f(x)]^2$. If $f(x) = x \ln x$ then $D^*f(x)\big|_{x=e}$ has the value
(A) e (B) 2e
(C) 4e (D) 8e

8. Suppose $I_1 = \int_0^{\pi/2} \cos(\pi \sin^2 x)dx$, $I_2 = \int_0^{\pi/2} \cos(2\pi \sin^2 x)dx$, $I_3 = \int_0^{\pi/2} \cos(\pi \sin x)dx$ then
(A) $I_1 = 0$ (B) $I_2 + I_3 = 0$
(C) $I_1 + I_2 + I_3 = 0$ (D) $I_2 = I_3$

Linked Comprehension Type

*This section contains 4 paragraphs. Based upon each paragraph, 2 multiple choice questions have to be answered. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.*

Paragraph for Question Nos. 9 to 10

Read the following write up carefully and answer the following questions:
Let the function f satisfies $f(x)\cdot f'(-x) = f(-x)\cdot f'(x)$ for all x and $f(0) = 3$

9. The value of $f(x)\cdot f(-x)$ for all x is
(A) 4 (B) 9
(C) 12 (D) 16

Space for rough work

10. $\int_{-51}^{51} \frac{dx}{3+f(x)}$ has the value equal to

(A) 17 (B) 34
(C) 102 (D) 0

Paragraph for Question Nos. 11 to 12

Read the following write up carefully and answer the following questions:
A curve is represented parametrically by the equations $x = e^t \cos t$ and $y = e^t \sin t$ where t is a parameter then

11. The relation between the parameter t and the angle α between the tangent to the given curve and the x-axis is given by 't' equals

(A) $\frac{\pi}{2} - \alpha$ (B) $\frac{\pi}{4} + \alpha$
(C) $\alpha - \frac{\pi}{4}$ (D) $\frac{\pi}{4} - \alpha$

12. If $f(t) = \int (x+y)dt$ then the value of $f\left(\frac{\pi}{2}\right) - f(0)$ is

(A) 1 (B) –1
(C) $e^{\pi/2}$ (D) 0

Paragraph for Question Nos. 13 to 14

Read the following write up carefully and answer the following questions:

A curve is represented parametrically by the equations $x = f(t) = a^{\ell n(b^t)}$ and $y = g(t) = b^{-\ell n(a^t)}$ a, b > 0 and $a \neq 1$, $b \neq 1$ where $t \in R$

13. Which of the following is not a correct expression for $\frac{dy}{dx}$?

(A) $-\frac{1}{f(t)^2}$ (B) $-(g(t))^2$
(C) $-\frac{g(t)}{f(t)}$ (D) $-\frac{f(t)}{g(t)}$

Space for rough work

14. The value of $\frac{d^2y}{dx^2}$ at the point where f(t) = g(t) is

(A) 0 (B) $\frac{1}{2}$

(C) 1 (D) 2

Paragraph for Question Nos. 15 to 16

Read the following write up carefully and answer the following questions:

Consider a quadratic function $f(x) = ax^2 + bx + c$ $(a, b, c \in R, a \neq 0)$ and satisfying the following conditions

(i) $f(x - 4) = f(2 - x)\ \forall\ x \in R$ and $f(x) \geq x\ \forall\ x \in R$

(ii) $f(x) \leq \left(\frac{x+1}{2}\right)^2\ \forall\ x \in (0, 2)$

(iii) The minimum value of f(x) is zero

15. The value of the leading coefficient to the quadratic polynomial is

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$

(C) $\frac{1}{2}$ (D) 1

16. f′(1) has the value equal to

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$

(C) $\frac{1}{2}$ (D) 1

Space for rough work

SECTION – A

Matching List Type

This section contains **4 multiple choice questions**. Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

17. Match the following List–I with List–II

List – I		List – II	
(P)	$\int \frac{x^2\left(x^6+x^5-1\right)}{\left(2x^6+3x^5+2\right)^2}\,dx$	1.	$\frac{x}{\left(x^2+4x+1\right)^{1/2}}+c$
(Q)	$\int \frac{\left(x^5+x^4+x^2\right)}{\sqrt{4x^7+5x^6+10x^4}}\,dx$	2.	$\frac{1}{2}\left(1+x^{-2}+x^{-5}\right)^{-2}+c$
(R)	$\int \frac{\left(2x^{12}+5x^9\right)}{\left(x^5+x^3+1\right)^3}\,dx$	3.	$-\frac{1}{6}\left(2x^3+3x^2+2x^{-3}\right)^{-1}+c$
(S)	$\int \frac{(2x+1)}{\left(x^2+4x+1\right)^{3/2}}dx$	4.	$x\left(\frac{x^3}{25}+\frac{x^2}{20}+\frac{1}{10}\right)^{1/2}+c$

Codes:

	P	Q	R	S
(A)	4	1	2	3
(B)	3	4	2	1
(C)	1	3	4	2
(D)	3	2	1	4

Space for rough work

18. Let $I=\int\frac{e^x}{e^{4x}+1}dx$ and $J=\int\frac{e^{-x}}{e^{-4x}+1}dx$. Then for any arbitrary constant C match the following List–I with List–II

List – I		List – II	
(P)	I	1.	$\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{e^{2x}-1}{\sqrt{2}-e^x}\right)+c$
(Q)	J + I	2.	$\frac{1}{2\sqrt{2}}\ln\left(\frac{e^{2x}-\sqrt{2}e^x+1}{e^{2x}+\sqrt{2}e^x+1}\right)+c$
(R)	J – I	3.	$\frac{1}{2\sqrt{2}}\left(\tan^{-1}\left(\frac{e^{2x}-1}{\sqrt{2}e^x}\right)-\frac{1}{2}\ln\left(\frac{e^{2x}-\sqrt{2}e^x+1}{e^{2x}+\sqrt{2}e^x+1}\right)\right)+c$
(S)	J	4.	$\frac{1}{2\sqrt{2}}\left(\tan^{-1}\left(\frac{e^{2x}-1}{\sqrt{2}e^x}\right)+\frac{1}{2}\ln\left(\frac{e^{2x}-\sqrt{2}e^x+1}{e^{2x}+\sqrt{2}e^x+1}\right)\right)+c$

Codes:

	P	Q	R	S
(A)	4	1	2	3
(B)	3	4	1	2
(C)	1	3	4	2
(D)	3	1	2	4

19. Match the following List–I with List–II

List – I		List – II	
(P)	If $f(x)=\int_0^{g(x)}\frac{dt}{\sqrt{1+t^3}}$ where $g(x)=\int_0^{\cos x}\left(1+\sin t^2\right)dt$ then the value of $f'\left(\frac{\pi}{2}\right)$	1.	3
(Q)	If f(x) is non-zero differentiable function such that $\int_0^x f(t)dt=\left(f(x)\right)^2$ for all x then 2 + f(2) equals	2.	2
(R)	If $\int_a^b\left(2+x-x^2\right)dx$ is maximum then (a + b) has the value equal to	3.	1
(S)	If $\lim_{x\to0}\left(\frac{\sin 2x}{x^3}+a+\frac{b}{x^2}\right)=0$ then (3a + b) has the value equal to	4.	–1

Codes:

	P	Q	R	S
(A)	4	1	2	3
(B)	3	4	1	2
(C)	4	1	3	2
(D)	2	1	4	3

Space for rough work

20. Match the following List–I with List–II.

	List – I		List – II
(P)	If $I=\int_2^3\left((x-1)^3+(4-x)^3+x\right)\cos\pi x dx$ then $\lvert 50\,\pi^2\, I\rvert$ is equal to	1.	0
(Q)	If $J=\int_0^{10}\text{sgn}(\sin\pi x)\,dx$ then 10J is equal to where sgn x denotes signam function of x	2.	100
(R)	If $K=\int_0^{102}\left[\cot^{-1}x\right]dx$ then 70 + [K] is equal to (where [y] denotes greatest integer equal to or less than y	3.	50
(S)	If $L=\frac{\int_0^{51}[x+25]dx}{\int_0^{51}\{x+25\}dx}$ then $\frac{L}{2}$ is equal to (where [y] and {y} denotes greatest integer function and fractional part of function	4.	70

Codes:

	P	Q	R	S
(A)	4	1	2	3
(B)	3	4	1	2
(C)	2	1	4	3
(D)	2	1	3	4

Space for rough work